WEEK8

Aim:

To develop a simple **Mini Web Application using Node.js and Express.js** with registration, login, dashboard and JSON file storage.

Description:

Develop a **Mini Web Application using Node.js and Express.js** based on a real-world application scenario. The application consists of a **Home/Landing Page, Registration Page, Login Page, and User Dashboard**. Users can register by entering their details, and the registration data is sent to the Express.js backend using a **POST request** and stored in a **JSON file** as a simple database. During login, the entered username and password are checked against the stored user details. If the credentials are valid, the user is allowed to access the dashboard; otherwise, an error message is displayed. The application also provides a **logout option** to return to the login page.

Program Code:

1) app.js:

```
const express = require("express");
const fs = require("fs");
const path = require("path");

const app = express();
const PORT = 3000;

app.use(express.urlencoded({ extended: true }));
app.use(express.json());

app.use(express.static(path.join(__dirname, "public")));

// Home
app.get("/", (req, res) => {
 res.sendFile(path.join(__dirname, "public", "index.html"));
});

// Register page
app.get("/register", (req, res) => {
 res.sendFile(path.join(__dirname, "public", "register.html")); });

// Register user
```

```
app.post("/register", (req, res) => {

 const newUser = {
 name: req.body.name,
 age: req.body.age,
 dob: req.body.dob,
 gender: req.body.gender,
 email: req.body.email,
 mobile: req.body.mobile,
 username: req.body.username,
 password: req.body.password,
 address: req.body.address
 };

 fs.readFile("users.json", "utf8", (err, data) => {  let users = [];

 if (!err && data.trim() !== "") {
 users = JSON.parse(data);
 }

 users.push(newUser);

 fs.writeFile(
 "users.json",
 JSON.stringify(users, null, 2),
 (err) => {
 if (err) {
 res.send("Error saving data");
 } else {
 res.send(`
 <h2>Registration Successful!</h2>  <a href="/login">Go to
Login</a>  `);
 }
 }
 );
 });
});

// Login page
app.get("/login", (req, res) => {
 res.sendFile(path.join(__dirname, "public", "login.html")); });

// Login
app.post("/login", (req, res) => {
```

```
const username = req.body.username;
const password = req.body.password;

fs.readFile("users.json", "utf8", (err, data) => {

if (err) {
res.send("Error reading data");
return;
}

const users = JSON.parse(data);

const user = users.find(
u => u.username === username &&
u.password === password
);
if (user) {
res.sendFile(
path.join(__dirname, "public", "dashboard.html")  );
} else {
res.send(`
<h2>Invalid Username or Password</h2>
<a href="/login">Try Again</a>
`);
}
});
});

// Logout
app.get("/logout", (req, res) => {
res.redirect("/login");
});

// Start server
app.listen(PORT, () => {
console.log(`Server running at http://localhost:${PORT}`); });
```

2)index.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Mini Web Application</title> <link

rel="stylesheet" href="style.css"> </head>

<body>

    <header>

        <h1>Mini Web Application</h1>

        <nav>

            <a href="/">Home</a>

            <a href="/register">Register</a>

            <a href="/login">Login</a>

        </nav>

    </header>

<main class="container">

        <h2>Welcome to Our Application</h2> <p>

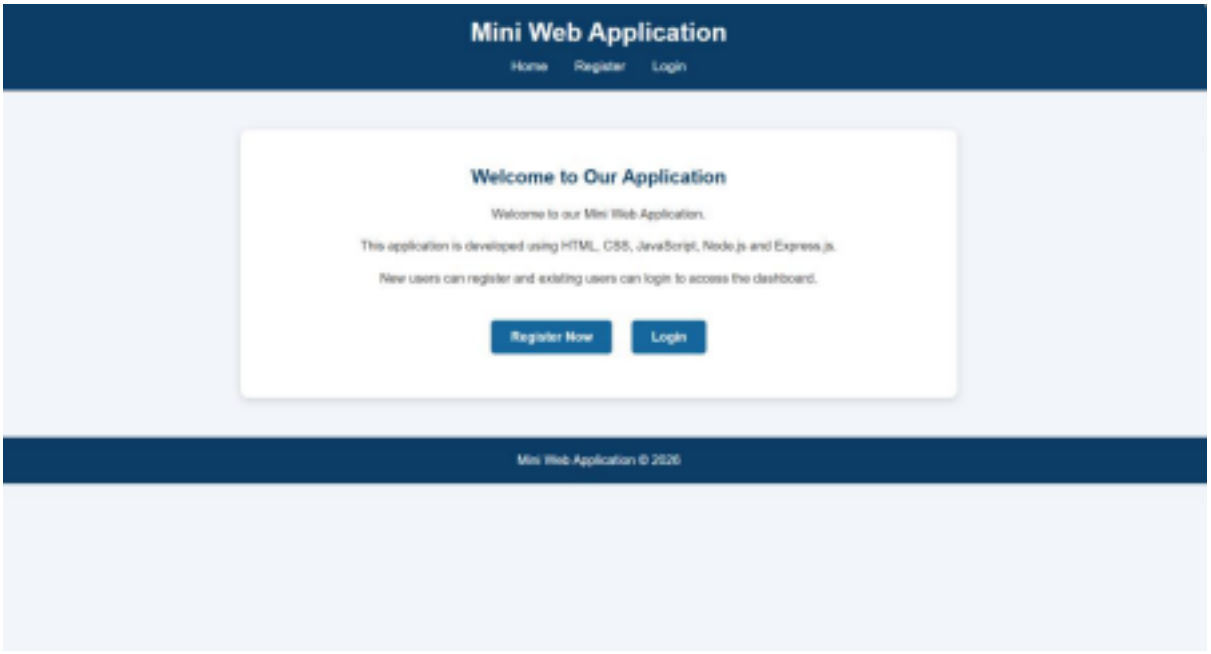
            Welcome to our Mini Web Application. </p>

        <p>

            This application is developed using
```

```
            HTML, CSS, JavaScript, Node.js and Express.js. </p>
        <p>
            New users can register and existing users
            can login to access the dashboard.
        </p>
        <div class="buttons">
            <a href="/register" class="btn">Register Now</a> <a
            href="/login" class="btn">Login</a>
        </div>
    </main>
    <footer>
        <p>Mini Web Application &copy; 2026</p> </footer>
</body>
</html>
```

Output:



3)register.html:

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
          content="width=device-width, initial-scale=1.0">

    <title>User Registration</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

    <header>
        <h1>User Registration</h1>

        <nav>

            <a href="/">Home</a>
```

```
        <a href="/register">Register</a>

        <a href="/login">Login</a>

    </nav>

</header>

<main class="form-container">

    <h2>Create Your Account</h2>

    <form action="/register" method="POST">

        <label>Name:</label>

        <input
            type="text"
            name="name"
            placeholder="Enter your name"
            required
        >

        <label>Age:</label>

        <input
            type="number"
            name="age"
            placeholder="Enter your age"
            required
        >

        <label>Date of Birth:</label>
        <input
            type="date"
            name="dob"
            required
        >

        <label>Gender:</label>

        <select name="gender" required>

            <option value="">
```

```
        Select Gender
    </option>

    <option value="Male">
        Male
    </option>

    <option value="Female">
        Female
    </option>

    <option value="Other">
        Other
    </option>

</select>

<label>Email:</label>

<input
    type="email"
    name="email"
    placeholder="Enter your email"
    required
>

<label>Mobile Number:</label>
<input
    type="tel"
    name="mobile"
    placeholder="Enter mobile number" required
>

<label>Username:</label>

<input
    type="text"
    name="username"
    placeholder="Create username"
    required
>

<label>Password:</label>

<input
```

```
                type="password"
                name="password"
                placeholder="Create password"
                required
            >

            <label>Address:</label>

            <textarea
                name="address"
                rows="4"
                placeholder="Enter your address"
                required></textarea>

            <button type="submit">
                Register
            </button>

        </form>
        <p class="bottom-text">

            Already have an account?

            <a href="/login">
                Login here
            </a>

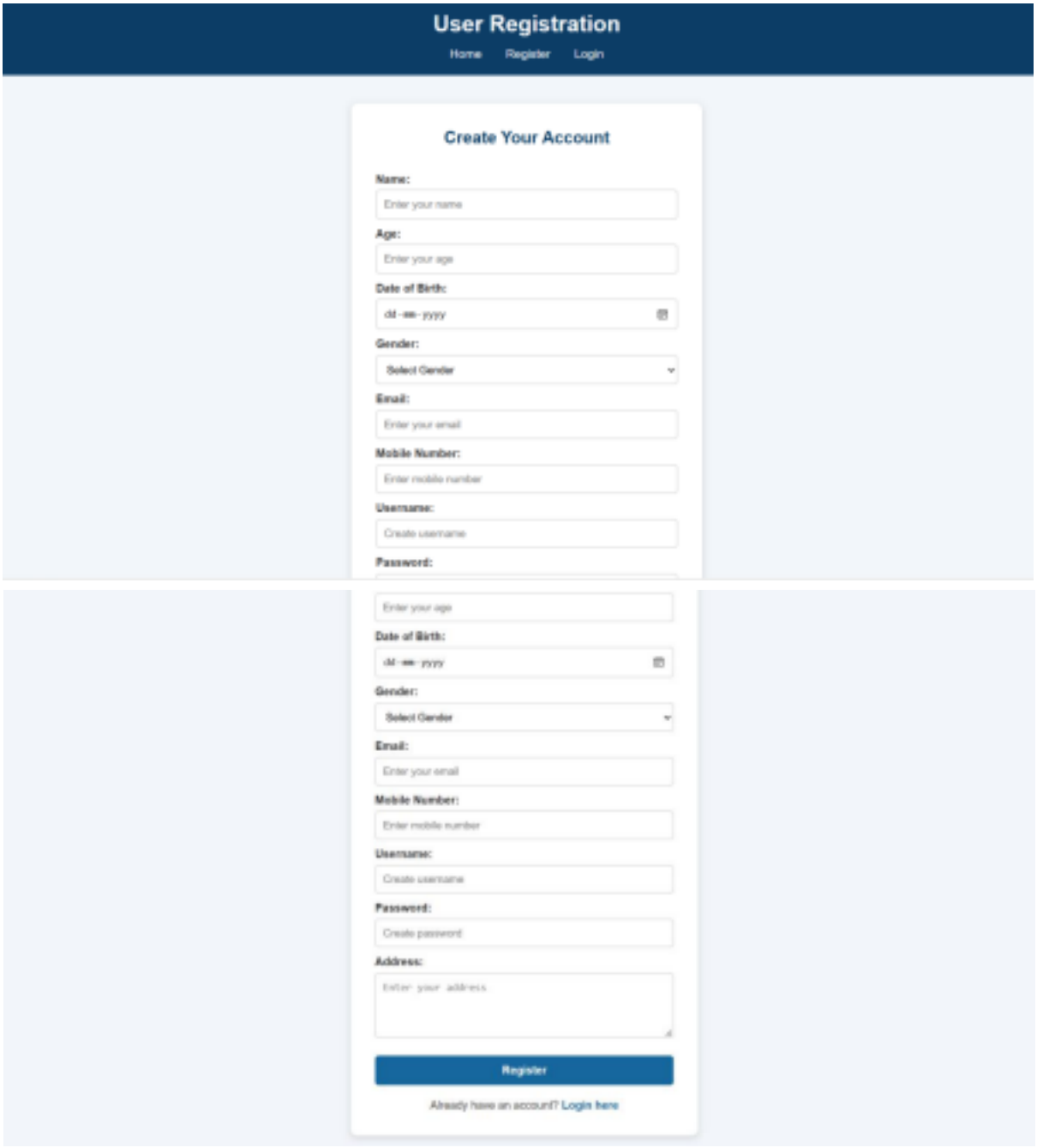
        </p>

    </main>

</body>

</html>
```

Output:



4)login.html:

```
<!DOCTYPE html>
<html>

<head>
 <title>Login</title>
 <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

<header>
 <h1>User Login</h1>

 <nav>
 <a href="/">Home</a>
 <a href="/register">Register</a>
 <a href="/login">Login</a>
 </nav>
</header>

<div class="form-container">
 <h2>Login</h2>

 <form action="/login" method="POST">

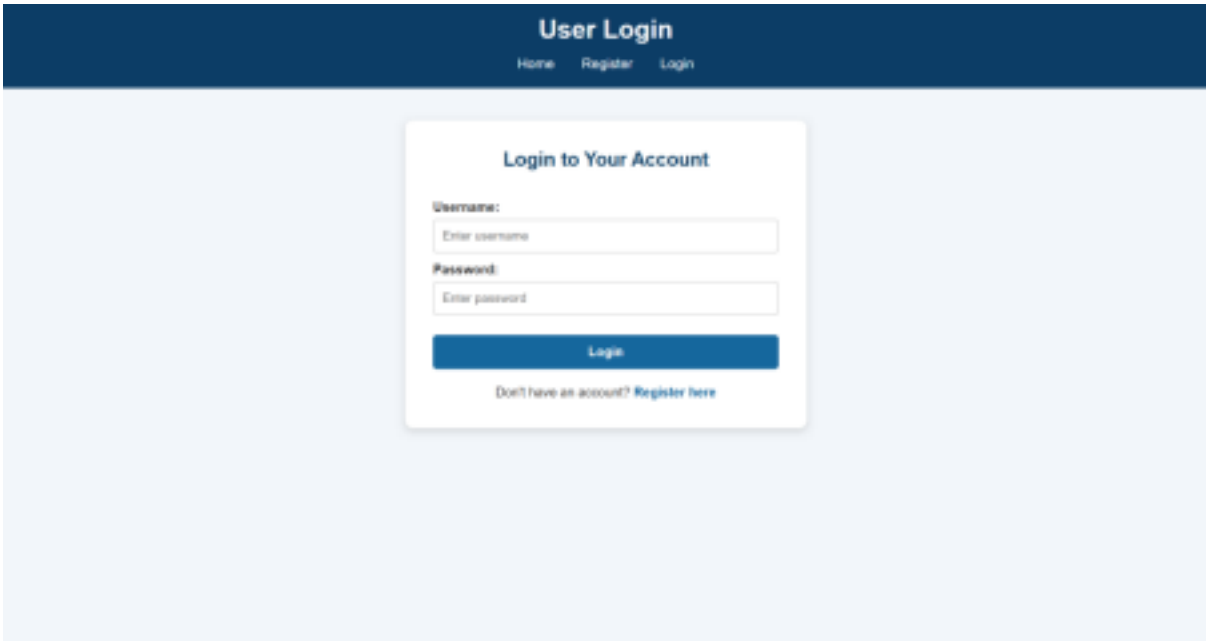
 <label>Username:</label>
 <input type="text" name="username" required>

 <label>Password:</label>
 <input type="password" name="password" required>

 <button type="submit">Login</button>

 </form>
 <p>
 Don't have an account?
 <a href="/register">Register</a>
 </p>
</div>
</body>
</html>
```

Output:



Style.css:

```
* {
 box-sizing: border-box; }

body {
 font-family: Arial;
 margin: 0;
 background-color: #f2f6fa; }

header {
 background-color: #0c3d66;  color:
white;
 text-align: center;
 padding: 20px;
}

nav a {
 color: white;
 text-decoration: none;
margin: 15px;
}

 .container {
 width: 80%;
 margin: 50px auto;
```

```
 padding: 40px;
 background-color: white;
text-align: center;
 border-radius: 10px;
}

.form-container {
 width: 500px;
 max-width: 90%;
 margin: 40px auto;
 padding: 30px;
 background-color: white;
border-radius: 10px;
}

form {
 display: flex;
 flex-direction: column; }

label {
 margin-top: 10px;
 margin-bottom: 5px;
 font-weight: bold;
}

input,
select,
textarea {
 padding: 10px;
 border: 1px solid #ccc;
border-radius: 5px;
}

button,
.btn {
 margin-top: 20px;
 padding: 12px;
 background-color: #15679d;  color:
white;
 border: none;
 border-radius: 5px;
 text-decoration: none;  cursor:
pointer;
```

```
}
button:hover,
.btn:hover {
 background-color: #0c3d66;
}
```

Users.json:

```
[]
```

Result: Thus, the Mini Web Application using Node.js and Express.js was successfully developed and executed. The application allows users to register, store their details in a JSON file, login using valid credentials, access the dashboard, and logout.